KALAM ([illegible]) — AGRICULTURAL INSIGHTS ATLAS

Ambasamudram & Cheranmahadevi Taluks, Tirunelveli · Multi-Year EO Analytics (2023–2026)

CROPPING INTENSITY

12.6% Double-Cropped Cropland

AUDIT READINESS

205 Targeted Audit Parcels (GPS)

ANNUAL WATER DEMAND

348.51 Million m³ (MCM) Canal Requirement

EXECUTIVE INSIGHT SUMMARY & POLICY ACTION ITEMS

• **Cropping Intensity:**	12.6% of cropland (5288.1 ha) in Ambasamudram & Cheranmahadevi is double-cropped with dual Kar and Pishanam cycles, alongside 0.9% under perennial banana canopy.
• **Transplanting Calendar:**	Peak Kar transplanting occurs in May (46.6% of Kar fields), while main Pishanam sowing peaks in Nov (39.5% of Pishanam fields). 0.0% of Pishanam acreage experiences delayed transplanting beyond November.
• **Field Verification Priority:**	25.0% of parcels (511 fields) are flagged for targeted ground audit due to lower phenological confidence or inter-annual crop instability, with top-50 priority targets packaged for GPS navigation.
• **Village-Level Statistics:**	Kallidaikurichi leads overall agricultural cultivation across both crops with 614.4 ha of paddy (68% share) and 7.7 ha of banana.
• **Irrigation Dependency & Water Access:**	8.9% of paddy acreage is concentrated within 500 m of a river channel, canal or irrigation tank. Banana accounts for 0.8% of cropland in the immediate 0–250 m corridor, decaying sharply beyond 1 km.
• **Crop Change & Farmland Dynamics:**	Evaluated 41,879.6 ha of 200,961.8 ha cropland (confidence ≥ 0.60): 26.3% of paddy and 0.8% of banana parcels remained stable across 2023–2026, with 705.9 ha (1.7%) identified as candidate paddy-to-banana diversification.
• **Crop Health & NDVI Anomaly:**	95.6% of analyzed cropland displays normal or above-normal vegetative vigour. Vegetative moisture/vigour stress is observed across 4.4% (1101.0 ha) of cultivated area.
• **Indicative Gross Irrigation Requirement:**	Total indicative gross irrigation requirement across analyzed farmed plains is 348.5 MCM/year. Paddy cycles (Kar & Pishanam) account for 75.0% (261.4 MCM), while perennial banana requires 7.4 MCM (2.1%).
• **Segment Size Distribution (Image-Derived):**	19.8% of image-derived field segments (9.9% of agricultural area) are below 1.0 ha. The mean segment size is 1.64 ha with a median of 1.41 ha.

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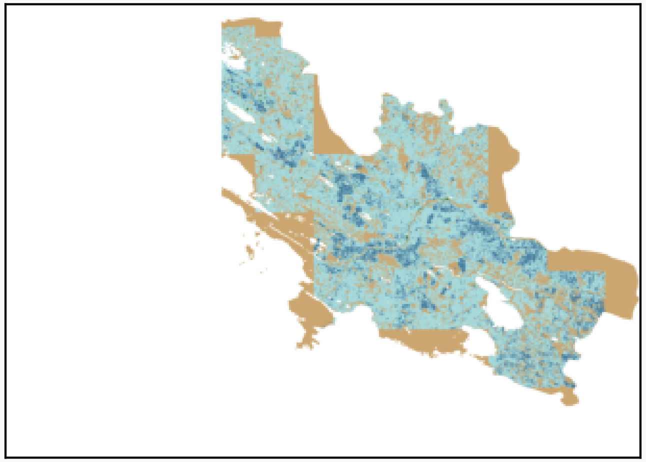
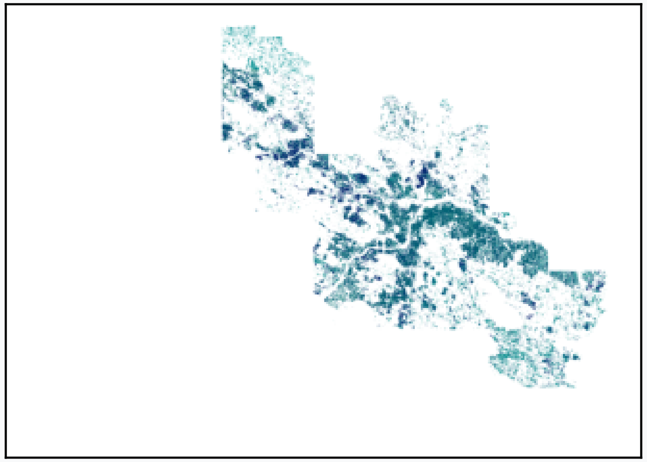
PHENOLOGY DYNAMICS & VERIFICATION PRIORITIZATION

01. Cropping Intensity



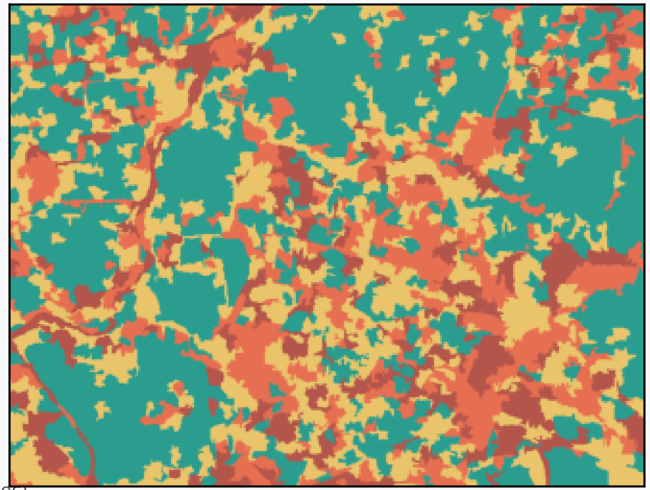
vated (7972.0 ha, 19.0%) | Single Crop (28259.8 ha, 67.5%) | Double Crop (5288.1 ha, 12.6%) | Triple Crop (2.3 ha, 0.0%)

02. Transplanting Calendar



Kar Peak (Jun–Jul) | Kar Late (Aug) | Pishanam Peak (Oct) | Pishanam Late (Nov–Dec)

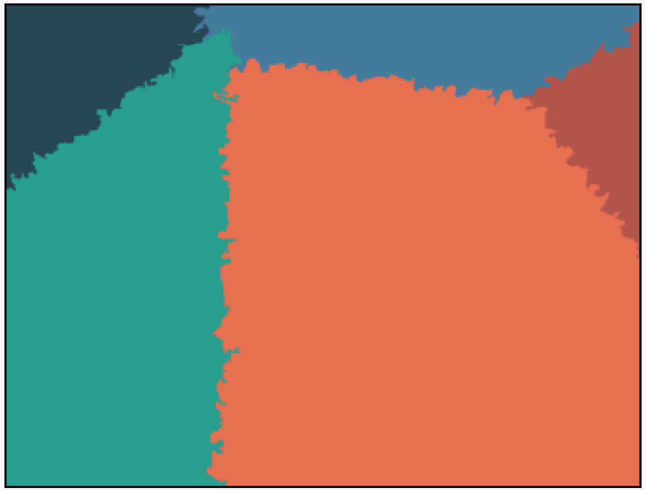
03. Field Audit Priority



Critical Priority (Top 10%) | High Priority (Next 15%) | Medium Priority (Next 25%) | Low Priority (Baseline 50

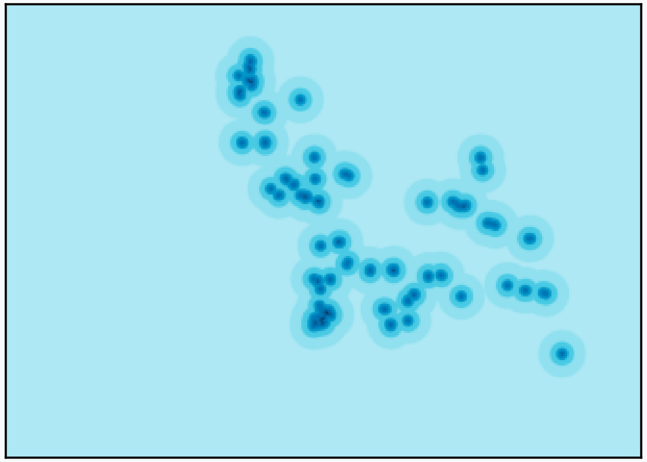
CANAL BUFFER DEPENDENCY & MULTI-YEAR CROP TRANSITIONS

04. Village Crop Allocations



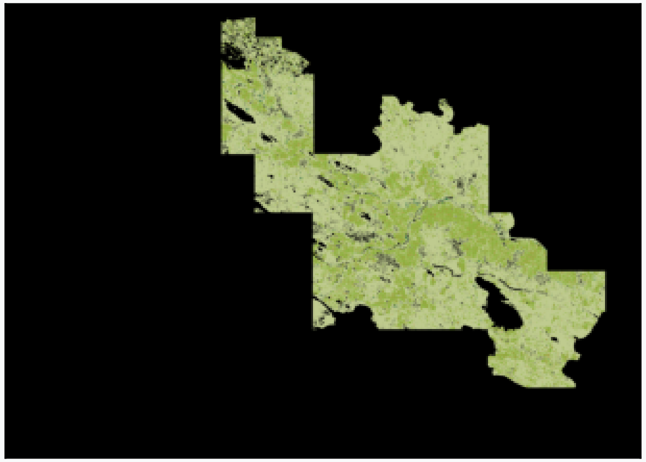
Urban (Paddy intensive) | Kallidaikurichi (Paddy intensive) | Veeravanallur (Diversified / Fallow) | Mukkudal (Paddy intensive)

05. Canal Buffer Decay



0 – 250 m | 250 – 500 m | 500 – 1,000 m | 1,000 – 2,000 m

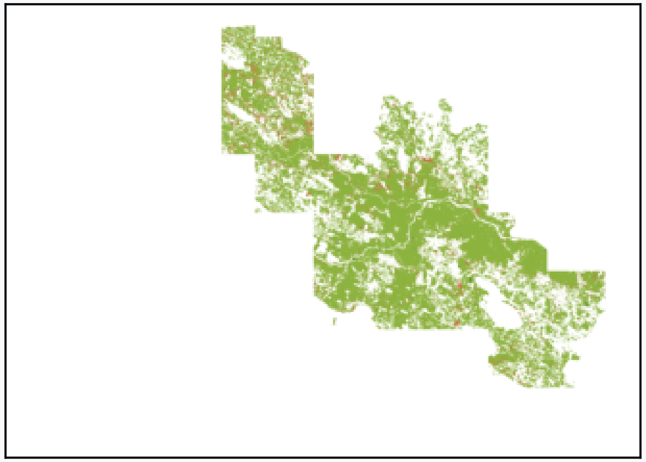
06. Crop Transitions (23–26)



Stable Paddy | Stable Banana | Candidate Paddy → Banana | Banana → Paddy

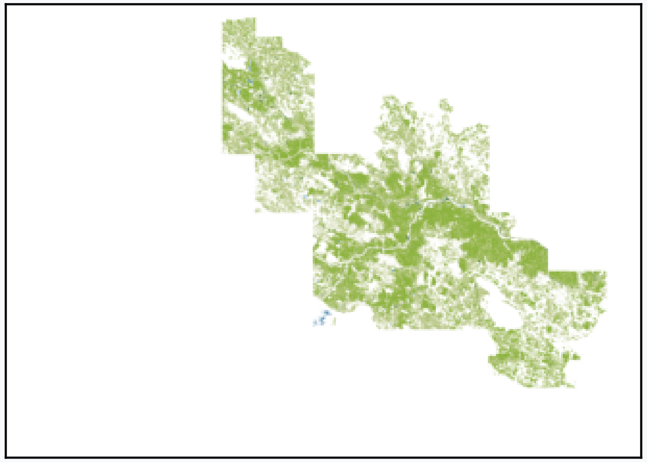
CROP VIGOUR, WATER DEMAND & PARCEL MORPHOLOGY

07. NDVI Vigour Anomaly



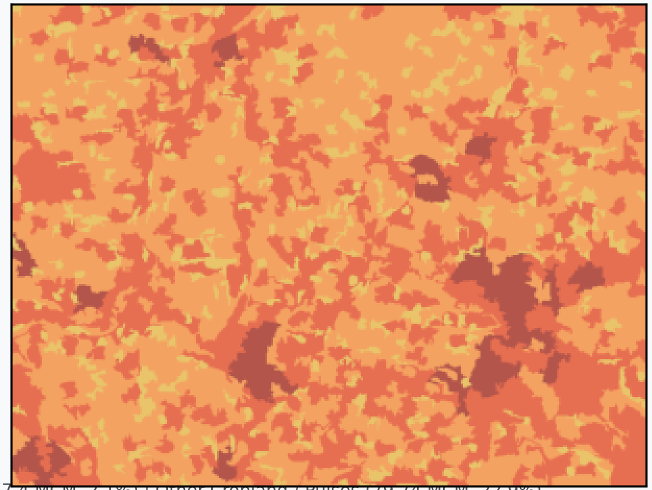
ss (186.8 ha, 0.7%) | Stressed (914.2 ha, 3.7%) | Normal Vigour (23,153.2 ha, 95.2%) | Above Normal (93.0 ha, 0.4%)

08. Water Demand Budget



Kar Paddy (Short-duration) (117.76 MCM, 33.8%) | Pishanam Paddy (Main Monsoon) (143.61 MCM, 41.2%) | Perennial Banana (Year-round) (7.4 MCM, 2.1%) | Other Cropland / Pulses (79.74 MCM, 22.9%)

09. Smallholder Fragmentation



Marginal (< 0.5 ha) (0.0% area, 0.0% count) | Small (0.5 – 1.0 ha) (9.9% area, 19.8% count) | Semi-Medium (1.0 – 2.0 ha) (50.6% area, 58.1% count) | M